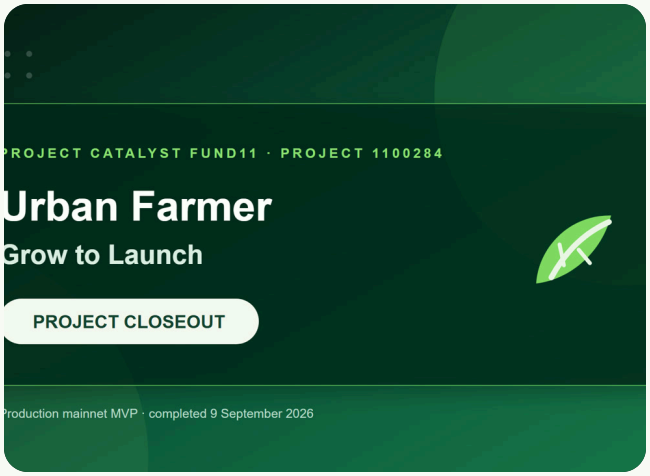
# Urban Farmer — Grow to Launch

Production farm management, crop education, and verifiable Cardano evidence

**EXECUTIVE SUMMARY**

Urban Farmer completed a production farming-management MVP configured for Cardano mainnet. The system combines wallet-signed authentication, RESI-gated operations, farm and season workflows, append-only Cropchain records, and season-level Merkle-root anchoring with ten professionally produced indoor-crop guides. Public transaction evidence, a repeatable validator, a production walkthrough, and this report make the final delivery independently reviewable.




PROJECT NAME
**Urban Farmer — Grow to Launch**

PROJECT MANAGER
**Lloyd Duhon**

PROJECT DATES
**March 11, 2024 – September 9, 2026**

OFFICIAL PROJECT URL
**https://milestones.projectcatalyst.io/projects/1100284**

PROJECT NUMBER / CATEGORY
**1100284 · Cardano Use Cases: Solution**

APPROVED FUNDING
**293,332 ADA**

FINAL RELEASE
**v1.0.2-mainnet · fa83073**

| Mainnet | 10 | 4 | 2 |
|---|---|---|---|
| PRODUCTION MVP | PUBLISHED CROP GUIDES | RECORDED FEEDBACK SESSIONS | AIKEN VALIDATORS |

## Challenge outcomes and delivery

| OUTCOME | RESULT | EVIDENCE-BASED EVALUATION |
|---|---|---|
| Functional production MVP | DELIVERED | Public dApp with wallet authentication, RESI access, farm/rack management, season operations, Cropchain Explorer, and Cardano anchoring. |
| Technical feasibility | CONFIRMED | Cash Register and Season Beacon actions are independently visible on Cardano mainnet; closed-season evidence validates against metadata label 674. |
| Progression from preprod | COMPLETED | The accepted preproduction architecture was retained and moved to the permanent production URL and mainnet configuration. |
| Education and engagement | DELIVERED | Ten professional crop guides are published; four recorded farmer/non-farmer feedback sessions informed presentation and instructional improvements. |
| Original 15-guide aspiration | 10 PRODUCED | Ten long-form guides were completed. This was the only original aspiration not fully realized; accepted milestone deliverables were completed. |

**Reporting boundary:** This report documents delivered outputs and technical verification. It intentionally excludes post-launch end-user counts, wallet/account totals, farm-activity totals, transaction-volume totals, and audience-view statistics.

## Production delivery and mainnet evidence

| DELIVERED COMPONENT | RESULT | PUBLIC EVIDENCE |
|---|---|---|
| Production application | DELIVERED | urban-farm.resi.works · public health, status, payment configuration, and crop-catalog endpoints |
| Cash Register mainnet deployment | CONFIRMED | f6a29d60…d29a0b · block 13,920,126 |
| Season Beacon operation | CONFIRMED | Start 0b6f5613…aa2037f · finish f9f91de4…fe34b48 |
| Filmed production acceptance run | PASSED | Start 09e0aafb…1dde23 · finish/anchor 51c112fb…8b276 |
| Independent anchor validation | VERIFIED | Three ordered records reproduce the Merkle root and agree with label-674 metadata; public verifier returned **verified: true**. |
| Final production release | VERIFIED | Release **v1.0.2-mainnet**, source commit **fa83073**; final source commit reported as **6b0e091** for v1.0.1-mainnet before the documentation fix. |

## Milestone 4 accepted outputs

| OUTPUT | RESULT | FINAL EVIDENCE |
|---|---|---|
| Production contracts, listeners, Cropchain, and dApp | DELIVERED | Mainnet configuration, transactions, walkthrough, test results, and validator evidence are public. |
| Production website for testers | DELIVERED | Permanent URL, in-app documentation, and reviewer runbook; reviewers may request complimentary RESI or use the Cash Register. |
| Production demonstration video | DELIVERED | Public 03:05 mainnet production walkthrough |
| Four additional crop guides | DELIVERED | Radish, Sunflower, Butter Lettuce, and Amaranth are public on YouTube and inside the production catalog. |

## Key achievements and collaboration

### Practical records, verifiable seasons

Operational crop events stay in an append-only record stream. Closing a season freezes the record set and anchors its deterministic Merkle root to Cardano, preserving independent verification without placing every farm event directly on-chain.

### Reviewer-friendly public evidence

Transaction links, endpoint snapshots, record-order hashes, validator output, and narrated walkthroughs create a durable evidence chain that can be checked without relying on project-team assertions.

### Education informed by feedback

Farmer and non-farmer feedback improved instructional flow, demonstrations, pacing, audio consistency, color, and chaptering. Ten professional long-form crop guides are integrated into the crop workflow.

### Multidisciplinary delivery

Software architecture, full-stack development, farming expertise, studio production, on-screen instruction, editing, and graphics were coordinated across the project team and Living Greens Urban Farm.

**Acceptance-test qualification:** The filmed path passed. Farm/rack creation, full harvest-and-pack lifecycle, reconciliation retry, and a separate logout/login persistence cycle were not rerun in the short recording and remain transparently documented as extended-regression items.

### Key learnings

Public proof must be designed alongside implementation. Transaction submission and indexer visibility are separate states, so durable systems persist submitted hashes and reconcile idempotently. Agriculture, software, and media pipelines also require distinct planning rhythms.

### Post-funding roadmap

Maintain the production service; add professional indoor-crop guides; develop Living Greens classroom grow kits; let older students document and track grows; improve educator onboarding, permissions, reporting, and public validation; and pilot with classrooms and farmers.

**Scope disclosure.** The original proposal aspired to fifteen crop guides; ten were produced and published. This was the only original aspiration not fully realized. The accepted Statement of Milestones deliverables were completed. The approved preproduction architecture—Cropchain records, two Aiken validators, and Cardano Merkle anchoring—was retained for mainnet.

**FDA horizon:** The FDA says the original Food Traceability Rule compliance date was January 20, 2026, proposed a 30-month extension to July 20, 2028, and states that Congress directed it not to enforce the rule before July 20, 2028. Urban Farmer will use this preparation window for traceability and classroom pilots; the product is not represented as a compliance guarantee.

## Primary public project sources

**Production application**
https://urban-farm.resi.works/

**Project Catalyst record**
https://milestones.projectcatalyst.io/projects/1100284

**Public closeout video**
https://youtu.be/rsUF6AAVB28

**Mainnet production walkthrough**
https://youtu.be/oX0RIyBBnbs

**Complete public evidence repository**
https://github.com/LloydDuhon/dRep/tree/main/project-catalyst/1100284

**Mainnet test results**
https://github.com/LloydDuhon/dRep/blob/main/project-catalyst/1100284/evidence/03-mainnet-test-results.md

**Anchor validation report**
https://github.com/LloydDuhon/dRep/blob/main/project-catalyst/1100284/evidence/17-anchor-validation-report.md

**Reviewer runbook**
https://github.com/LloydDuhon/dRep/blob/main/project-catalyst/1100284/evidence/02-mainnet-reviewer-runbook.md

## Milestone 4 crop-guide evidence

01 **Radish Microgreens**
https://youtu.be/9MeYiMZ4XcY

02 **Sunflower Microgreens**
https://youtu.be/XjtoW53t5gQ

03 **Butter Lettuce**
https://youtu.be/oBAABR3QPiQ

04 **Amaranth Microgreens**
https://youtu.be/QqFseKMfW1M

**Completion statement: Urban Farmer completed its accepted milestones on September 9, 2026, with a public production mainnet MVP, ten professional crop guides, independently inspectable Cardano evidence, a public closeout report, and a public closeout video.**

FDA source: FDA Food Traceability Rule · Project Catalyst closeout guidance: Project Close-out Report and Video

# Supplemental Clarifications Requested During Completion Review

At Project Catalyst's request, this section documents the status of several concepts referenced in the original voted proposal that were not separately defined as outputs, acceptance criteria, or evidence requirements in the subsequently reviewed and approved Statement of Milestones.

These clarifications supplement the historical project record. They do not modify the completion status of the accepted milestones, which were assessed through their respective Proof of Achievement reviews and Project Catalyst signoffs.

## Relationship between the voted proposal and the accepted milestones

The original voted proposal initiated the project's funding approval and established its broad direction, intended benefits, and preliminary delivery plan. Following the vote, the project entered the Fund 11 onboarding and milestone-development process. Through that process, the proposal was converted into a detailed Statement of Milestones specifying the funded outputs, acceptance criteria, evidence requirements, delivery schedule, and milestone allocations.

The Statement of Milestones was reviewed and accepted by the assigned Milestone Reviewers and subsequently approved by Project Catalyst. That accepted Statement of Milestones became the operative delivery and assessment framework for the project. The project's performance and completion are therefore judged against those approved milestones and their associated Proofs of Achievement.

The additional information below is provided at Project Catalyst's request to reconcile concepts mentioned in the original voted proposal with the final milestones accepted through the formal onboarding process. It supplies historical context and records how those concepts were addressed, refined, superseded, or excluded when the definitive milestone scope was established. It does not create additional retrospective deliverables, alter the accepted milestone criteria, or change the completion status of milestones already reviewed and approved.

## Discovery and node tooling

Milestone 1 formalized the accepted node-tooling requirement as a Terraform demonstration deploying a Cosmos node and a client relayer/listener-indexer that monitored Cardano network events associated with the project's smart contract. The submitted demonstration showed the Terraform deployment, Cosmos-based components, and Cardano listener/indexer operating together. The Milestone 1 Proof of Achievement was subsequently approved.

The original proposal also discussed a broader discovery concept intended to help Cardano Stake Pool Operators identify and participate in sidechain opportunities. This was not established as a separate output, acceptance criterion, or evidence requirement in the approved Statement of Milestones.

The completion-review discussion addressed the status of this broader discovery concept and the market-demand considerations that informed the final accepted project scope.

## Cosmos-node tooling and Cardano integration

The Cosmos-node work was completed through the accepted Milestone 1 and Milestone 2 requirements.

Milestone 1 demonstrated Terraform deployment of the Cosmos node, supporting services, faucet, interface, and Cardano listener/indexer. Milestone 2 demonstrated the accepted integration workflow: tRESI was locked through a Cardano testnet smart contract, the listener detected the transaction, and the corresponding test voucher asset was minted and displayed in the matched Keplr sidechain wallet.

These Proofs of Achievement were reviewed and approved under the milestone process.

## Evolution from IOHK tooling to custom Aiken validators

The original proposal contemplated adapting the IOHK sidechain contract tooling. During subsequent development, the broader ecosystem's sidechain-development direction shifted toward Midnight. Within the Urban Farmer implementation, custom Aiken validators replaced the initially contemplated IOHK contract tooling.

The resulting Season Beacon and Cash Register validators, Cardano listeners, append-only Cropchain records, and Merkle-root anchoring architecture were documented in the Milestone 3 preproduction Proof of Achievement. That implementation received two reviewer approvals and Project Catalyst administrative signoff. The accepted preproduction architecture was then carried into the Milestone 4 production release, which also received two reviewer approvals and Project Catalyst administrative signoff.

## Source code and DAO disposition

The original proposal stated that the application source code would be provided to the DAO and that the DAO would determine its licensing.

The DAO's token holders subsequently voted to retain the application source code as private intellectual property. In accordance with that community decision, the application source code was not released publicly or transferred for open-source licensing. The project's public evidence, transaction records, deployment configuration, validation materials, and educational outputs remain publicly accessible.

## Block Bounty system

The short proposal summary referenced deployment of a bounty system. The Block Bounty concept was not included as an output, acceptance criterion, or evidence requirement in the approved Statement of Milestones.

Following limited community engagement with the concept, the team discontinued the Block Bounty system rather than continuing development of a feature for which sufficient participation had not materialized.

## Educational-video quantity

The original proposal narrative discussed educational material covering fifteen crops or planting types. During the Statement of Milestones process, the accepted educational deliverables were formalized as ten unique professionally produced grow guides:

- One Broccoli Microgreens guide developed and subsequently improved through user feedback.
- Five guides delivered through Milestone 3.
- Four additional guides delivered through Milestone 4.

The reduction from the original fifteen-video concept reflected the effect of ADA price fluctuations on the available production budget. All ten educational guides defined through the approved milestones were completed, published on YouTube, and incorporated into the Urban Farmer application.

## Impact measurement, market conditions, and transaction evidence

The original proposal identified user uptake and transaction volume as potential measures of longer-term impact. The approved Statement of Milestones did not establish a numerical user-adoption target, transaction-volume threshold, measurement period, or associated acceptance criterion.

The project nevertheless supplied public technical transaction evidence demonstrating the Cardano integration, including validator deployment, season-start transactions, season-closing transactions, metadata-label-674 anchors, and independent Merkle-root validation.

The timing of broader deployment trials and adoption measurement was affected by two external developments:

- The federal Food Traceability Rule enforcement timeline was extended to July 2028, reducing the immediate compliance pressure that had been expected to drive near-term adoption.
- Local extension-office funding reductions delayed planned deployment trials and related outreach opportunities.

Accordingly, the closeout evidence documents the completed product, working Cardano transactions, and technical verification without representing acceptance-test activity as independent end-user adoption.

## Reconciliation conclusion

This section completes the requested documentary reconciliation between the original voted proposal and the final accepted Statement of Milestones. The proposal served as the starting point for the project, while the subsequent milestone-development and approval process established the specific outputs, acceptance criteria, and evidence against which delivery would be evaluated.

The project completed those accepted milestone requirements and submitted the corresponding evidence through the Proof of Achievement process. The explanations above record the historical disposition of original-proposal concepts that were not separately incorporated into the final milestone requirements. They are provided for transparency and completeness and do not represent outstanding milestone obligations or modify the basis on which the project's completion is assessed.